DAY 182/365 · KNN · PART 2 OF 3

How near is near? And how many neighbours?

Distance, choosing K, and why scaling matters.

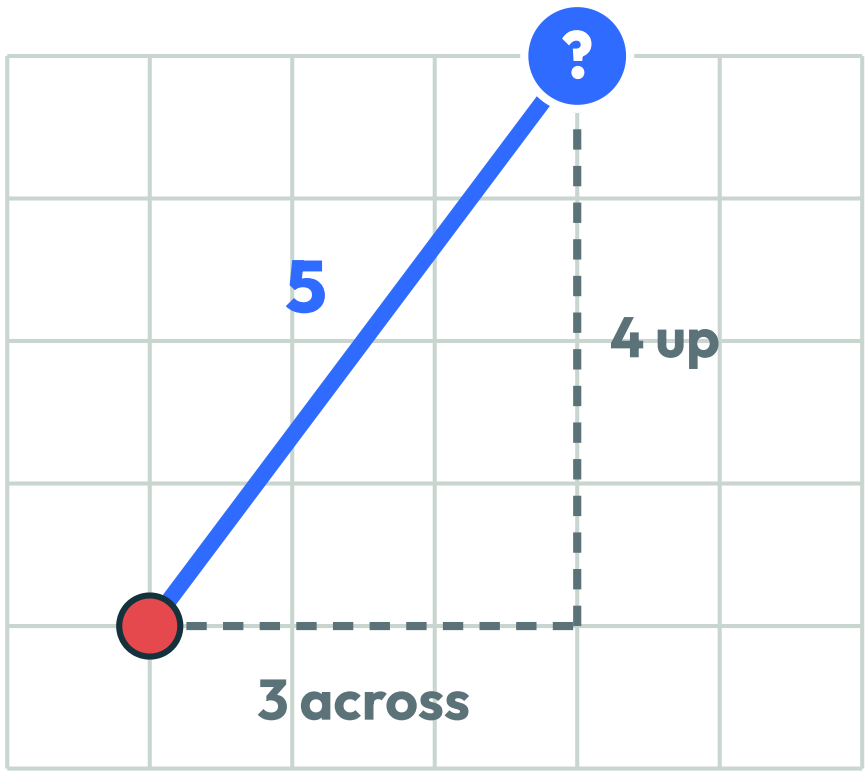


Suyog Yadav

Swipe →

Euclidean: as the crow flies

In Part 1 we measured the **straight line** between two points. Its proper name is **Euclidean distance**.



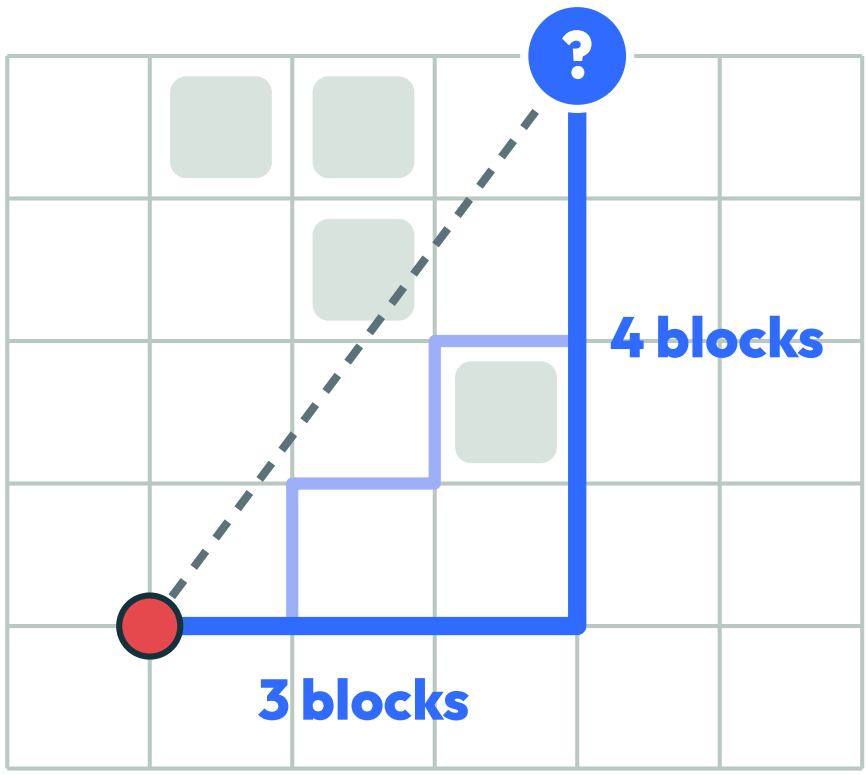
straight line
"as the crow flies"

$$\sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$$

Square each gap, add them up, take the square root. This is the most common choice.

Manhattan: as the taxi drives

In a city of blocks, you can't cut through buildings. You go along the streets.



taxi route: 7

every staircase: 7

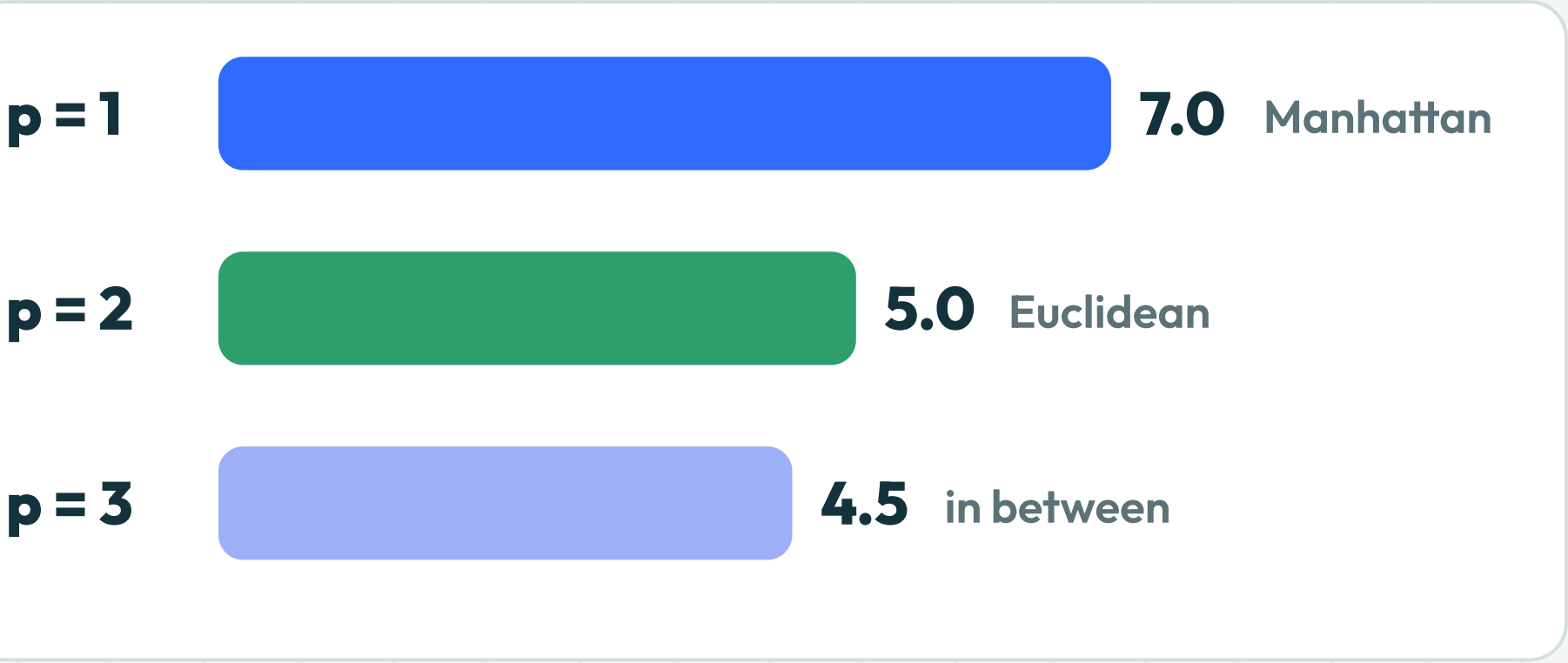
dashed line: 5
(buildings in the way)

|4 − 1| + |5 − 1| = 3 + 4 = 7

Just add up the gaps. Every staircase route has the same length.

Minkowski: one formula, a dial called p

Euclidean and Manhattan are two settings of the same family. The dial is **p**. Same two points, (1, 1) and (4, 5):



p = 1 gives Manhattan. **p = 2** gives Euclidean. Other values sit in between.

In scikit-learn, the default is Minkowski with p = 2, which is Euclidean. (My addition, not from the article.)

Units can fool the distance

Find the neighbour closest to someone aged **30** earning **₹50,000**:

	Neighbour 1 31 years, ₹58,000	**Neighbour 2** 60 years, ₹51,000
Raw distance	8,000.0	**1,000.4 ← nearest**
After scaling	**0.10 ← nearest**	0.75

What happened?

Raw, an income gap of ₹8,000 looks huge next to an age gap of 1 year, so **income drowns out age**. KNN picks a 60-year-old as the closest match to a 30-year-old.

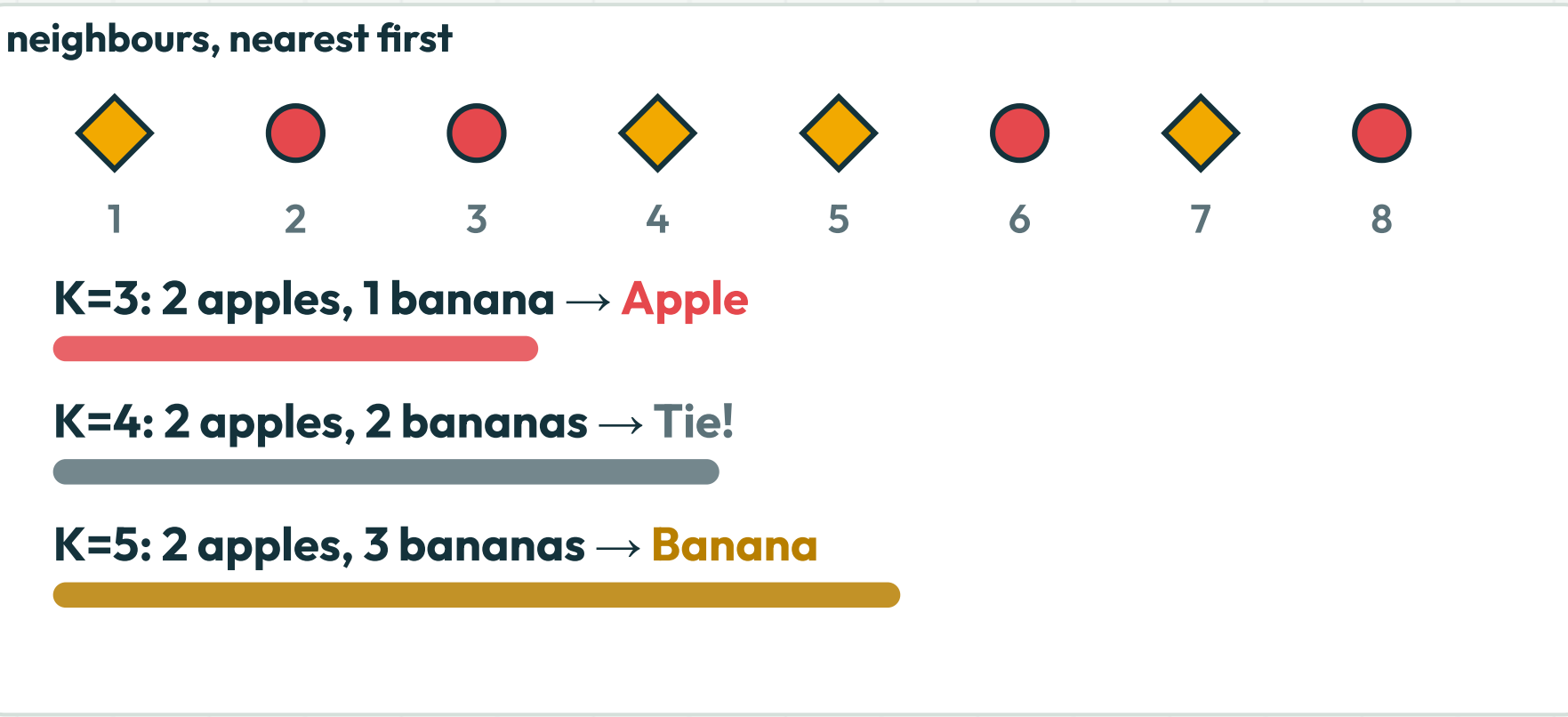
The fix

Put every column on a similar scale first. Here I divided each gap by its column's range (about 40 years, about ₹80,000).

Illustrative people and ranges. Same lesson as SVM Part 4.

Choosing K

Here are the 8 neighbours of the new point from Part 1, in order of closeness:



K too small

Listens to one or two neighbours. A single odd point can mislead it.

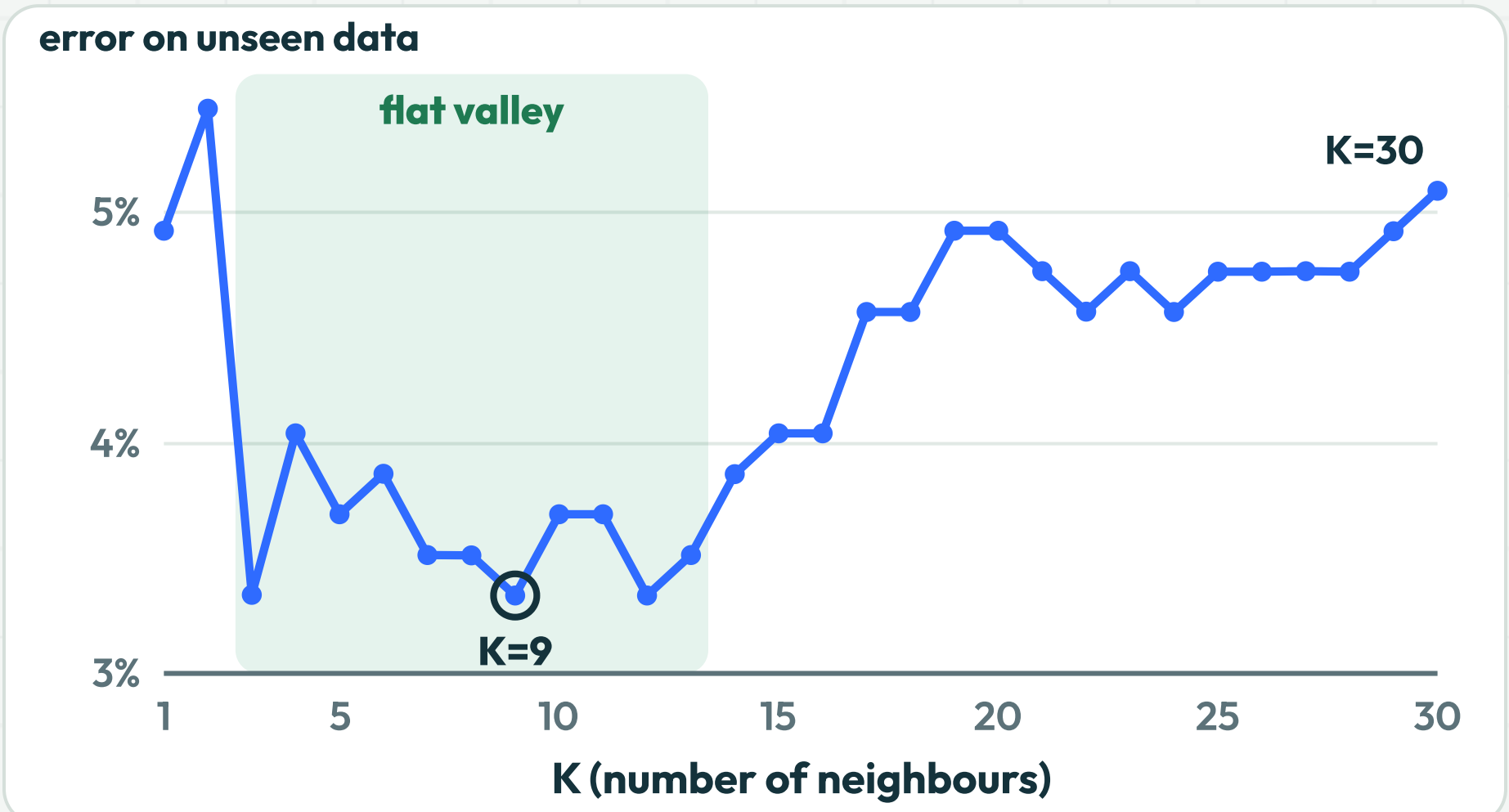
K too large

Listens to far-away points too, so the answer gets blurry.

Odd K avoids ties when there are two categories, as K = 4 just did.

Let the data pick K

Same method as choosing the polynomial degree on Day 167: test each K on data it hasn't seen (5-fold cross-validation) and plot the error. Real run, breast cancer data:



No sharp elbow, a flat valley. K = 1 gives **4.9%** error and K = 30 gives **5.1%**. Anything from 3 to 13 sits between 3.3% and 4.0%.

3.34% error at K = 3, 9 and 12 (tied). That's 2 patients out of 569 better than K = 5, so don't over-tune. I'd take K = 9: odd, and in the valley. Scaled data, scikit-learn 1.9.1, practice dataset.

TAKEAWAYS

Measure carefully, then let the data choose K.

Distance: Euclidean is a straight line, Manhattan is a taxi route, and Minkowski covers both with p.

Scale first, or the column with the biggest numbers decides everything.

K: too small is jumpy, too large is blurry. Try values with cross-validation and prefer odd.

Part 3: KNN in Python, from scratch and with scikit-learn, plus advantages and limitations.